Chapter 1: Introduction

1.1 Motivation

Currently, in the world of live electronic music performance, there is a tendency to use deep neural models to control timbre in a very expressive way. The human voice is a very expressive, natural and simple way to control such systems and does not need any special equipment except a microphone.

Neural audio synthesis is now advancing thanks to key models such as WaveNet (van den Oord et al., 2016), NSynth (Engel et al., 2017), and DDSP (Engel et al., 2020). However, these initial models are not very suitable for live use. Autoregressive models require a lot of computing power, and on the other hand, early differentiable signal processing models often have to focus on explicit tone tracking or processing in windows that can cause noticeable lag.

The launch of RAVE (Caillon & Esling, 2021) revolutionized the field by facilitating high-quality synthesis even on easily accessible CPUs to normal users. BRAVE (Caspe et al., 2025) goes a step further with a causal transmission architecture aimed at creating interactive and responsive music in real-time. However, converting voice to instrument is quite difficult when the source and destination timbres are very different acoustically. For example, plucked string instruments, such as the guitar, have a high-pitched percussion attack, and the sound decays exponentially, which is very different from the sustained, formant-rich structure of the human voice. This work investigates if causal transmission systems can achieve voice-to-guitar resynthesis, which sound perceptually convincing to human ears, with a latency of 10ms or less so it's usable in live performance. Solving this cross-modal mismatch is crucial to the enhancement of music interfaces that enable human vocal intuition to be the foundation for machine synthesis.

1.2 Objectives and Scope

To address the research question, this project defines a general objective to design and evaluate a streaming neural resynthesis pipeline for expressive voice-driven instrumental synthesis. This investigation is organized according to the following specific objectives:

- **Examine whether it is possible** to keep the total latency below 10 ms in a visual programming environment with a causal Variational Autoencoder (VAE) architecture.
- **Assess the performance** of instrument-specific models when driven by both vocal percussive inputs and sustained tonal singing.
- **Document training instabilities** and failure modes throughout the RAVE/BRAVE architecture family.
- **Evaluate the limits** of audio reconstruction fidelity when models are presented with out-of-distribution vocal signals.

The scope of this project is restricted to monophonic audio resynthesis since this constraint lowers computational complexity and also enables stable inference on consumer-grade devices. Consequently, polyphonic modeling, offline batch processing, and linguistic speech conversion (text-to-speech) are considered out-of-scope.

1.3 Methodology

The approach combined an iterative system implementation with structured diagnostic protocols. It was divided into five processes:

1. **Dataset preparation:** Public guitar and drum-kit corpora were preprocessed to a uniform 44.1 kHz mono format. This stage included a quantitative analysis of noise-floor heterogeneity to identify potential biases in the training data.
2. Training procedure: Models were trained under a two-step procedure: An independent representation learning phase to optimize a multi-scale STFT loss, then adversarial fine-tuning that is made to improve perceptual naturalness.
3. **Deployment:** The trained TorchScript models were integrated into Pure Data through the nn~ external, supporting live vocal interaction at a 64-sample block size.
4. **Quantitative Evaluation:** The main focus was using end-to-end latency benchmarks to evaluate the overall performance of the system. Furthermore, the degree of alignment between the input and output Short-Time Fourier Transform (STFT) magnitudes was also considered as a metric for spectral reconstruction closeness.
5. **Qualitative testing:** The author carried out listening tests to evaluate phonetic plausibility and flexibility for live vocal control.

1.4 Contributions

Deployment and assessment have brought about a number of technological and methodological advancements in neural audio synthesis:

- Voice-to-drums concept: A working prototype based on Pure Data, which shows an effective mapping control for percussive synthesis where vocal transients are converted to drum timbres with a latency of about 7 ms.
- Guitar autoencoder confirmation: A piece that accomplishes recognizable in-distribution reconstruction using the official RAVE v2 architecture within a causal framework, and it can be used as a reference point for cross-modal evaluation.
- Failure taxonomy: This is a systematic account of four different training collapse modes, amongst which are posterior collapse and GAN discriminator dominance, that have been witnessed in multiple training iterations.
- Dataset heterogeneity exploration: A numerically based investigation spotting noise floor variability in various public guitar datasets, which seem to be encoded in the latent space feature that is salient.
- Structural stabilization: An experimental proof that certain aspects of RAVE v2, for instance, a 4:1 generator-to-discriminator update ratio and multi-period discriminators (Kong et al., 2020), are vital in preventing adversarial mode collapse when operating under streaming constraints.

Such reflection could provide useful insights for further development of RAVE series architectures in the context of musical interaction that is responsive.

1.5 Document Structure

The thesis is organized into six chapters. Chapter 2 reviews the state of the art in neural synthesis and timbre transfer. Chapter 3 details the methodology, focusing on system architecture and training protocols. Chapter 4 describes the implementation of the Pure Data environment and the software pipeline. Chapter 5 presents the evaluation results, including the guitar failure taxonomy and the out-of-distribution analysis. Chapter 6 summarizes the findings and proposes future work regarding vocal preprocessing. Technical configurations and code are provided in the appendices.